

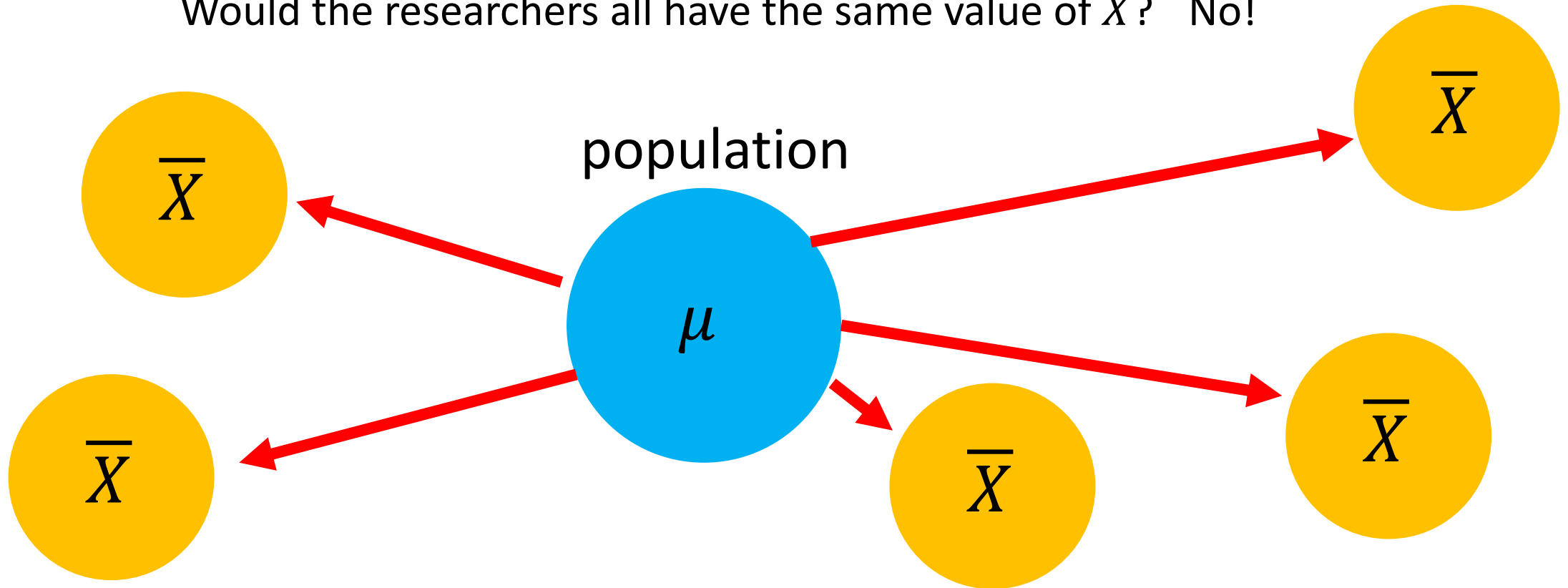
Unit 06 (B): Central Limit Theorem

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Suppose many researchers (independently of each other) each take a random sample from the same population and compute a sample mean.

Would the researchers know their value of $\bar{X}$ in advance? No!

Would the researchers all have the same value of $\bar{X}$? No!



The sample mean $\bar{X}$ is a random variable. And like all random variables, it has an expected value, variance, and standard deviation.

$$\begin{aligned} E(\bar{X}) &= \mu_{\bar{X}} \\ Var(\bar{X}) &= \sigma_{\bar{X}}^2 \\ SD(\bar{X}) &= \sigma_{\bar{X}} \end{aligned}$$

Some
courses/texts
use this
notation

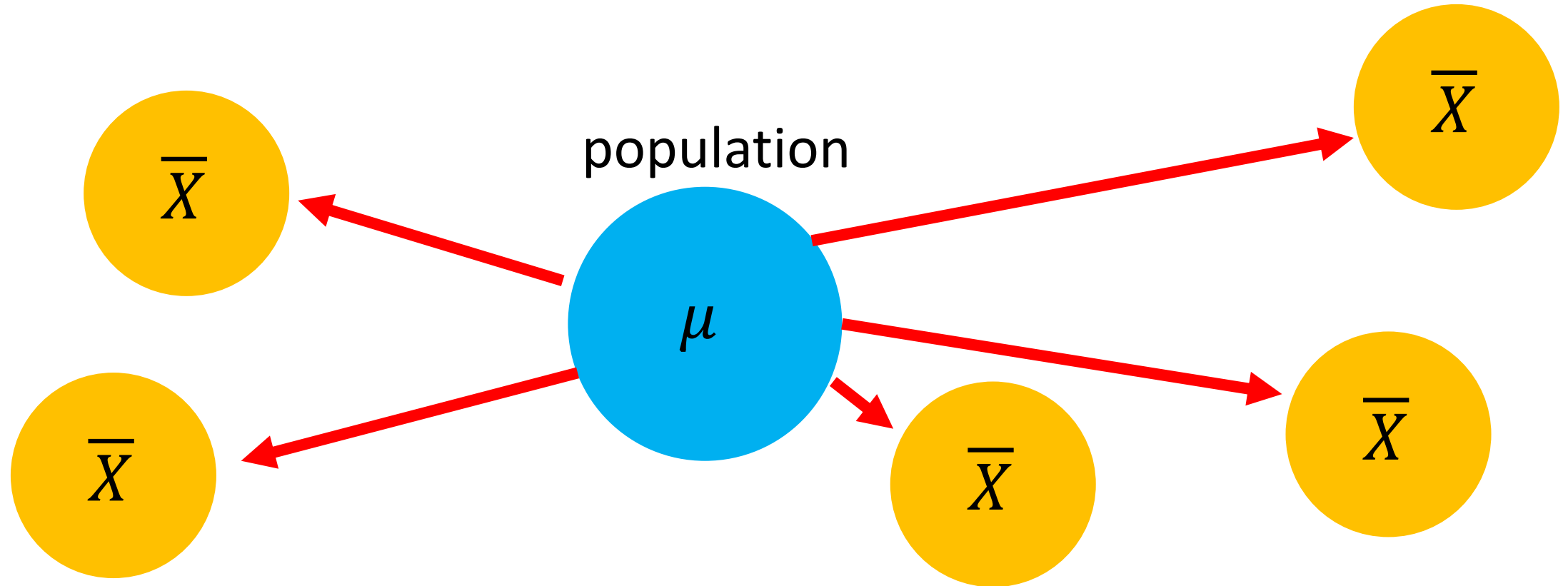
$X_1, \dots, X_n$ are i.i.d. random variables with $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2$.

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n} E(X_1 + X_2 + \dots + X_n) \\ &= \frac{1}{n} [E(X_1) + E(X_2) + \dots + E(X_n)] = \frac{1}{n} (\mu + \mu + \dots + \mu) \\ &= \frac{1}{n} (n\mu) = \mu \end{aligned}$$

$$E(\bar{X}) = \mu$$

The expected value of $\bar{X}$ equals μ . If the expected value of an estimator equals the parameter it estimates, we say it is unbiased.

Since $\bar{X}$ is unbiased for μ , if we were to find the average among all the $\bar{X}$'s (the mean among the sample means), it should be close to μ !



$X_1, \dots, X_n$ are i.i.d. random variables with $E(X_i) = \mu$, $\text{Var}(X_i) = \sigma^2$.

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n^2} \text{Var}(X_1 + X_2 + \dots + X_n)$$

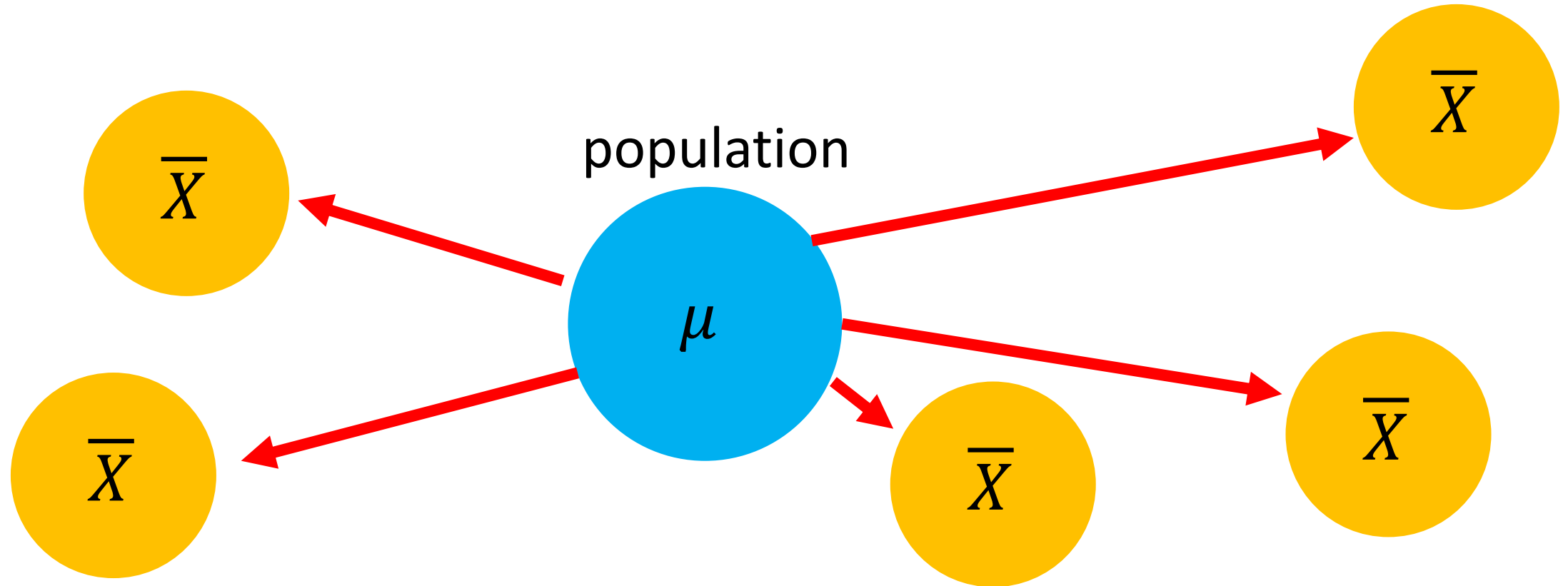
$$= \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)] = \frac{1}{n^2} (\sigma^2 + \sigma^2 + \dots + \sigma^2)$$

$$= \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}$$

$$\boxed{\text{Var}(\bar{X}) = \frac{\sigma^2}{n}} \longrightarrow \boxed{SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}}$$

Consider the set of all the $\bar{X}$'s.

What would we prefer: a lot of variation among the sample means,
or little variation?



We prefer unbiased estimators with a small variance & standard deviation.

$$E(\bar{X}) = \mu \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \quad SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

If we want these quantities to be small, we should have a large n . The larger the sample size, the more precision we can expect from $\bar{X}$ as an estimator.

In the previous video we found the mean, variance, and standard deviation of the sample mean.

$$E(\bar{X}) = \mu \quad Var(\bar{X}) = \frac{\sigma^2}{n} \quad SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Next question:

What kind of random variable is $\bar{X}$?

This question is answered by the most famous and important theorem in all of statistics, the

Central Limit Theorem

Central Limit Theorem Part 1

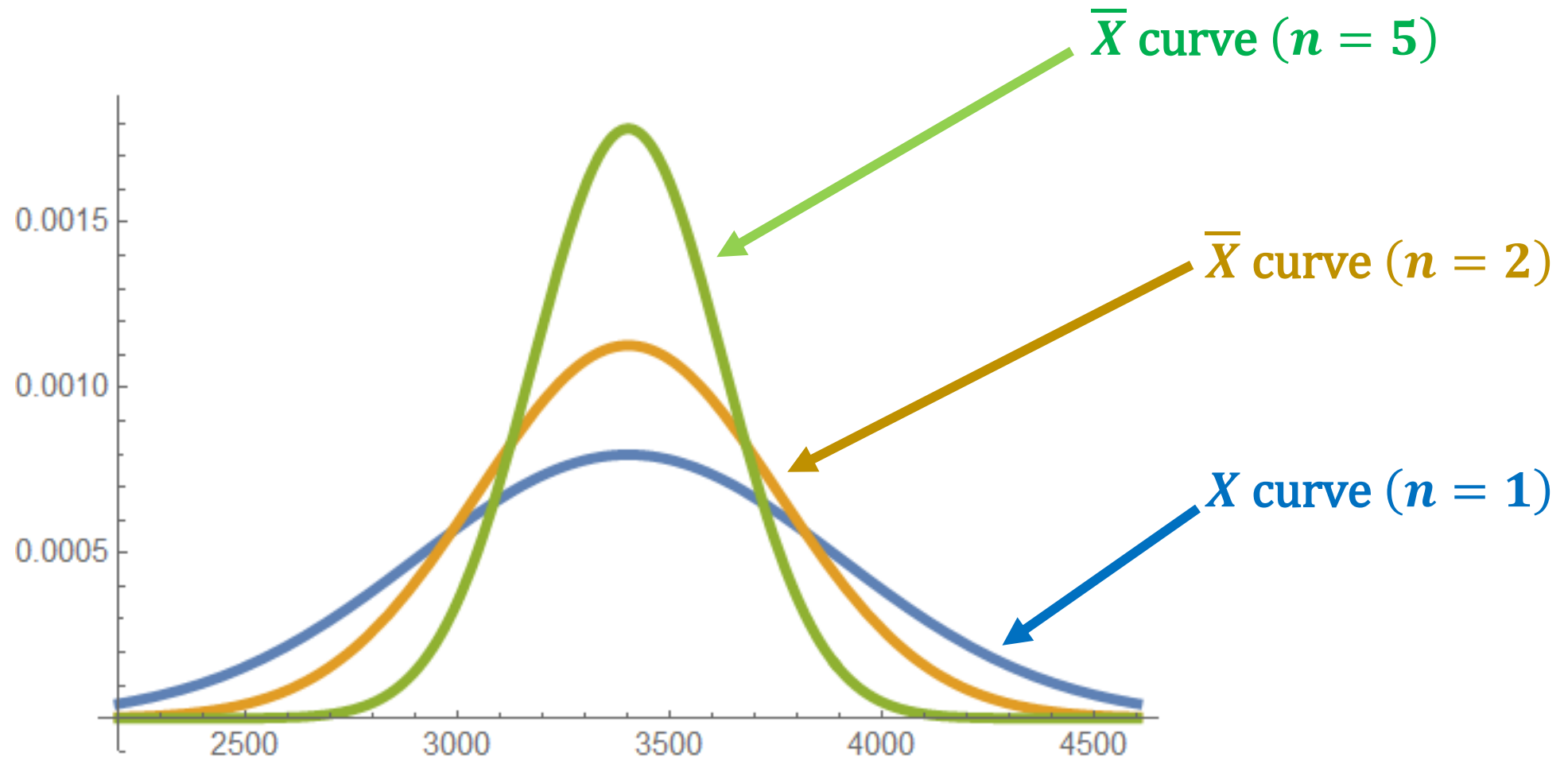
Suppose a random sample is taken from a normally distributed population.

$$X \sim N(\mu, \sigma)$$

Then the sample mean is also normally distributed:

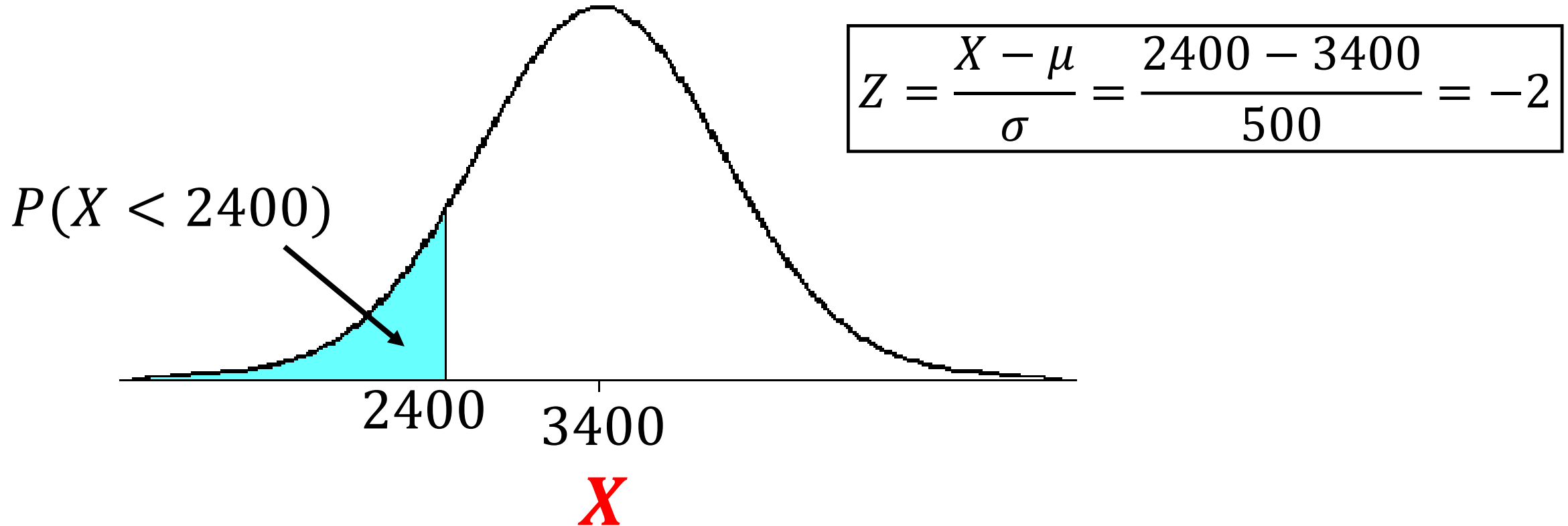
$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

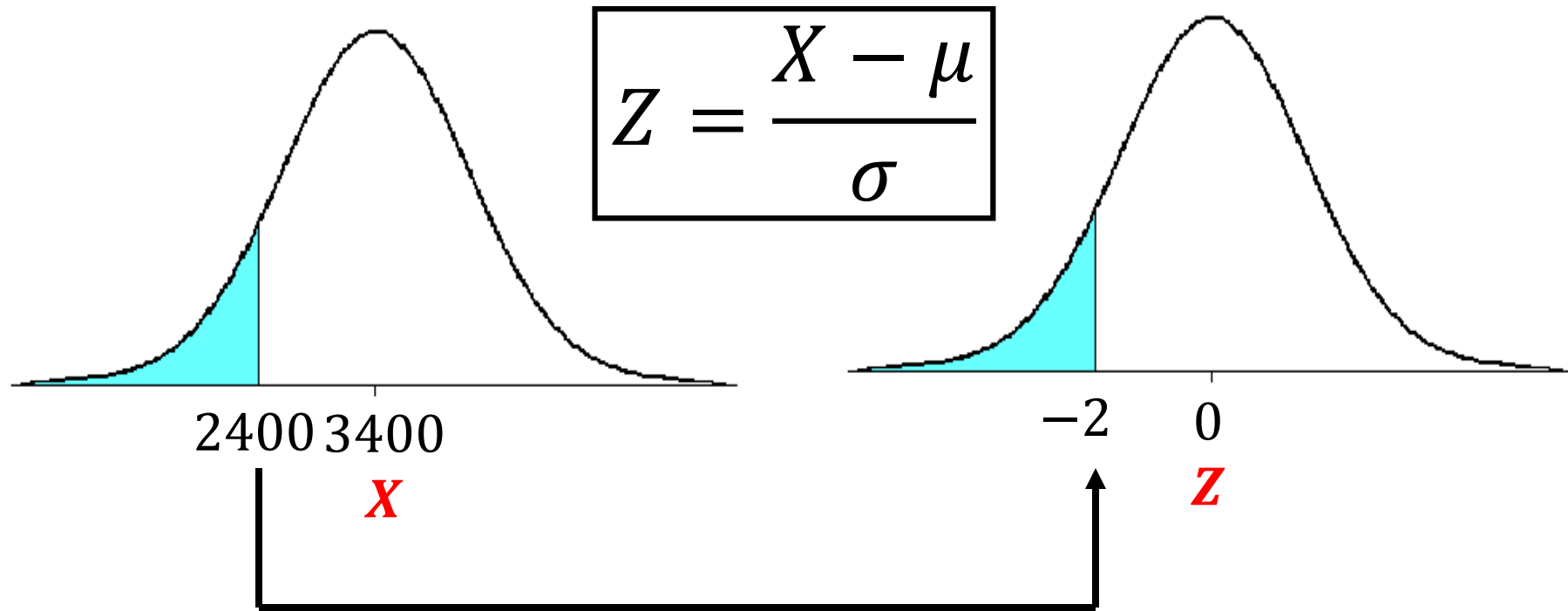
The $\bar{X}$ curve is centered around the same mean as the X curve, but is taller and thinner due to a smaller standard deviation.



X = mass of a randomly selected newborn baby

$X \sim N(\mu = 3400 \text{ grams}, \sigma = 500 \text{ grams})$





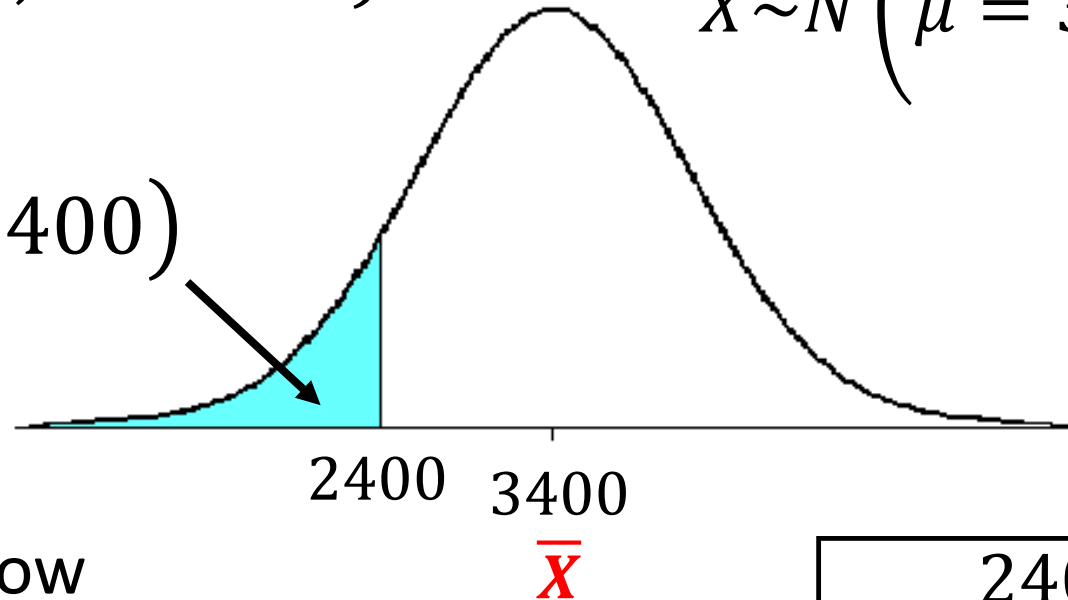
$$P(X < 2400) = P\left(Z < \frac{2400 - 3400}{500}\right) = P(Z < -2) = .0228$$

Instead of randomly selecting one baby, what if we randomly select 2 babies and compute the sample mean?

$$X \sim N(\mu = 3400, \sigma = 500)$$

$$\bar{X} \sim N\left(\mu = 3400, \frac{\sigma}{\sqrt{n}} = \frac{500}{\sqrt{2}}\right)$$

$$P(\bar{X} < 2400)$$



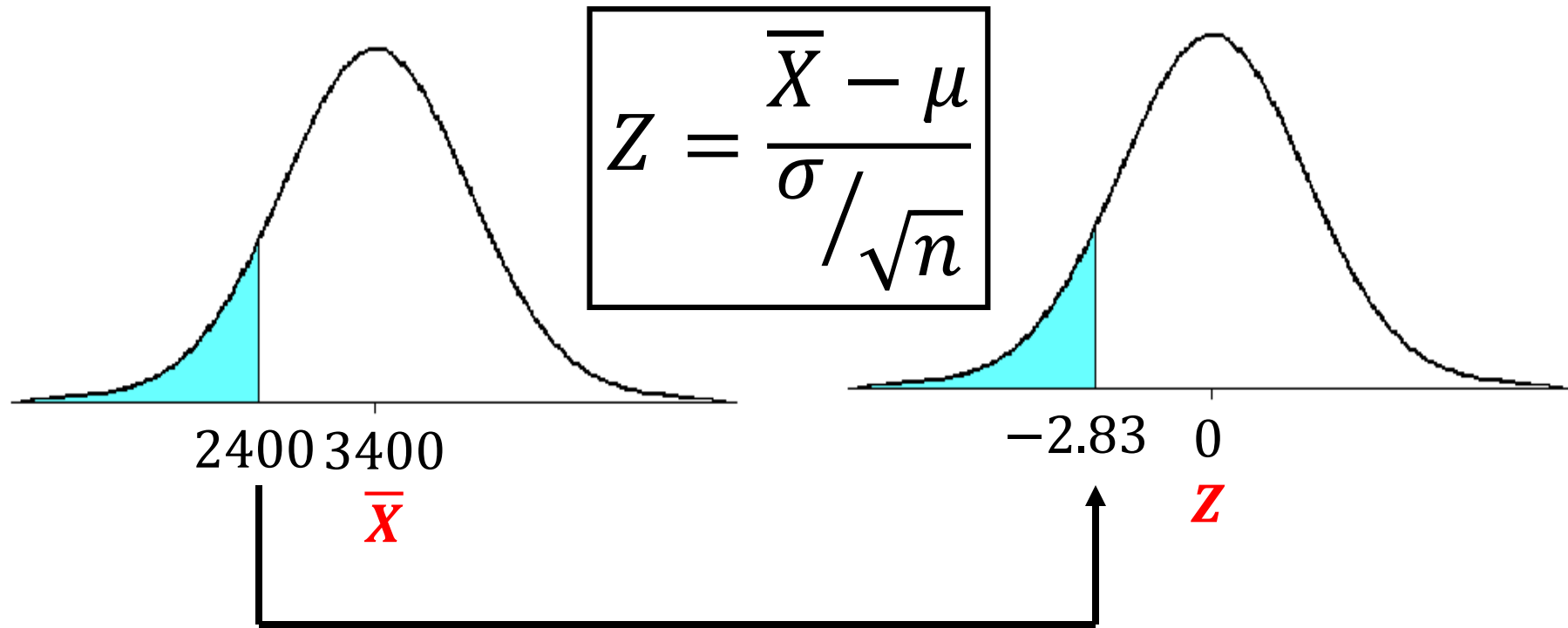
before

$$Z = \frac{X - \mu}{\sigma}$$

now

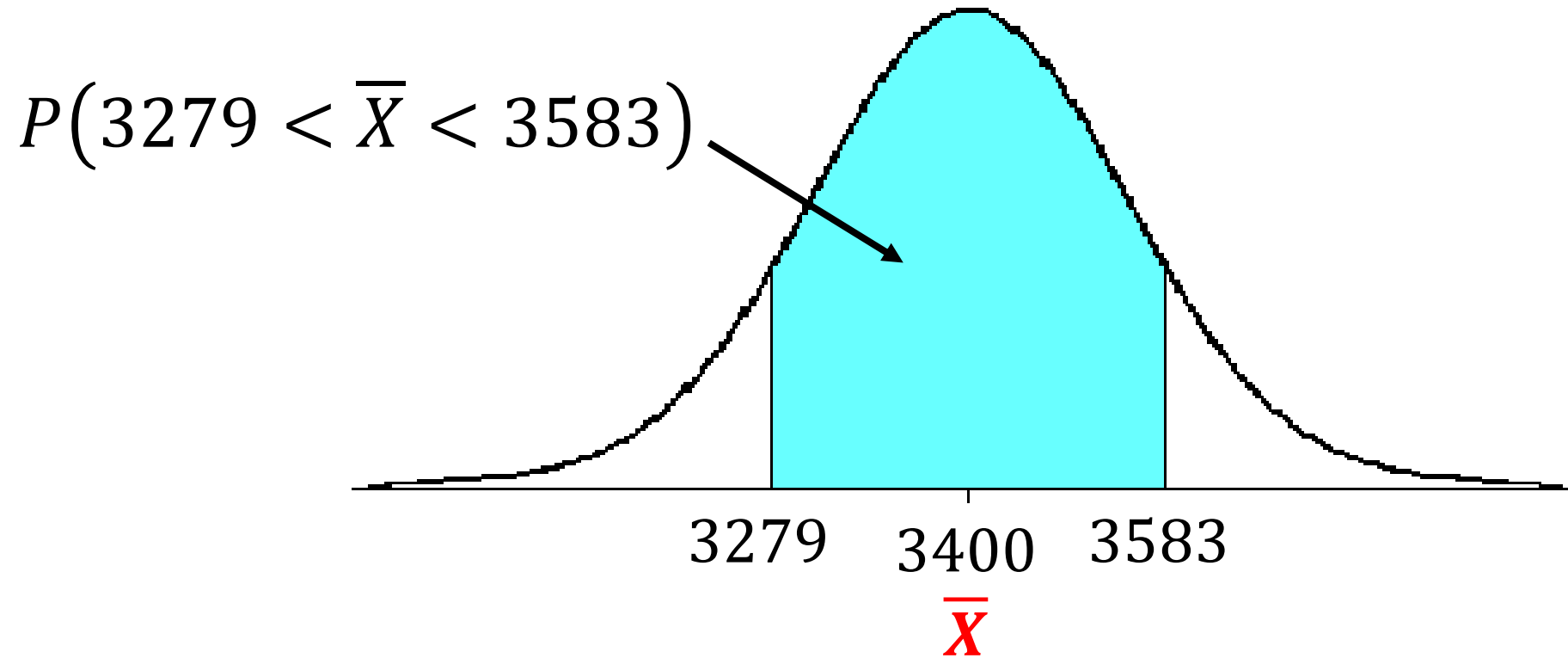
$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

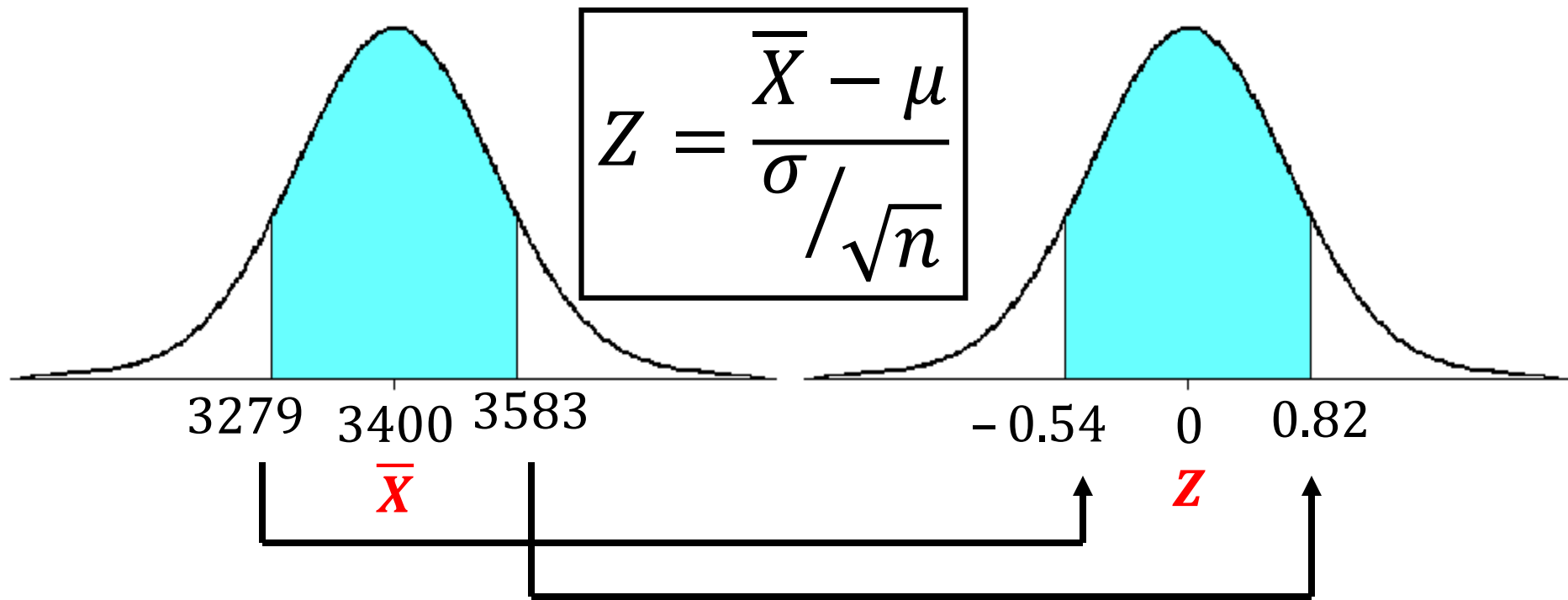
$$Z = \frac{2400 - 3400}{500 / \sqrt{2}} \approx -2.83$$



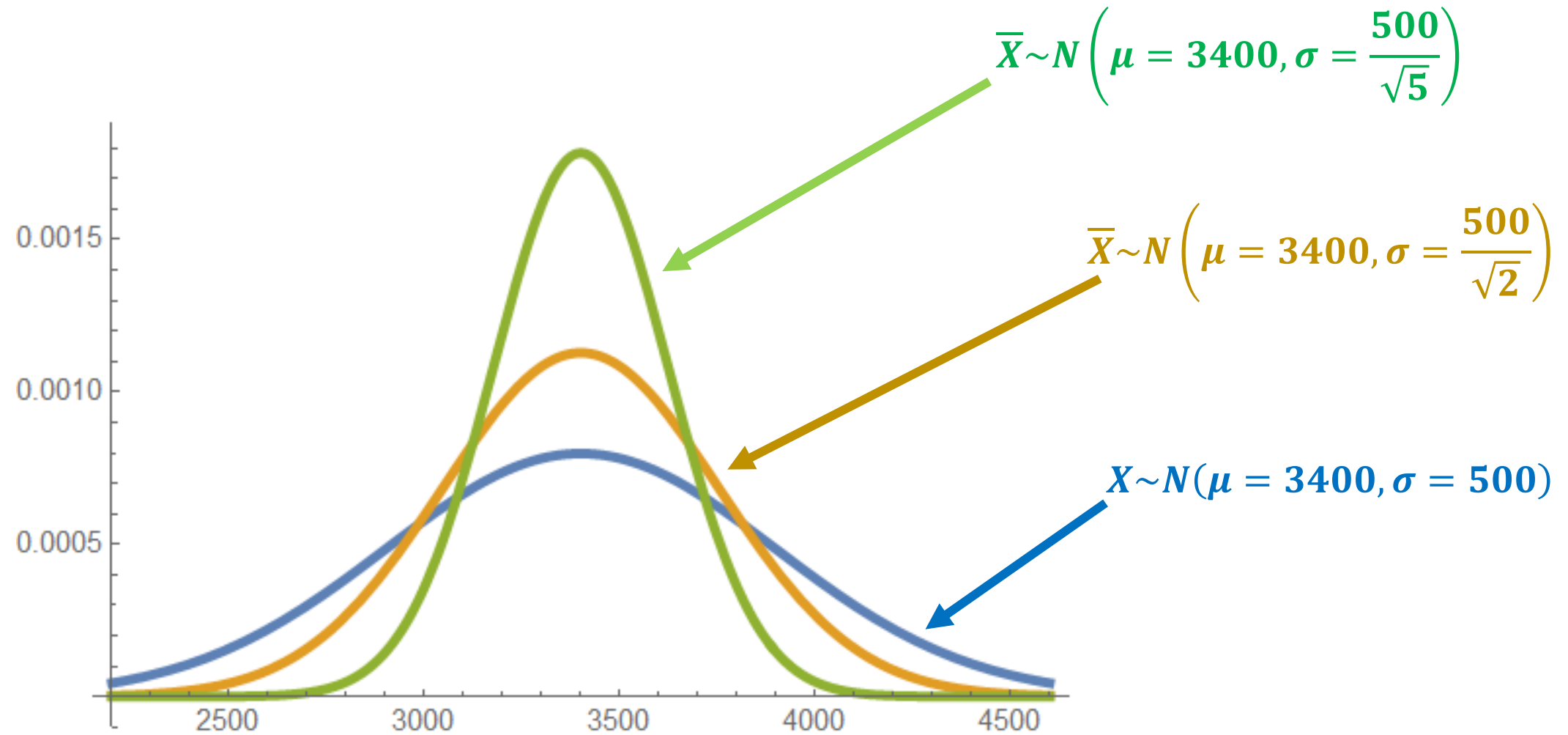
$$P(\bar{X} < 2400) = P\left(Z < \frac{2400 - 3400}{500 / \sqrt{2}}\right) = P(Z < -2.83) = .0023$$

What if we randomly select 5 babies?





$$\begin{aligned}
 P(3279 < \bar{X} < 3583) &= P\left(\frac{3279 - 3400}{500/\sqrt{5}} < Z < \frac{3583 - 3400}{500/\sqrt{5}}\right) = \\
 &= P(-0.54 < Z < 0.82) = .7939 - .2946 = .4993
 \end{aligned}$$



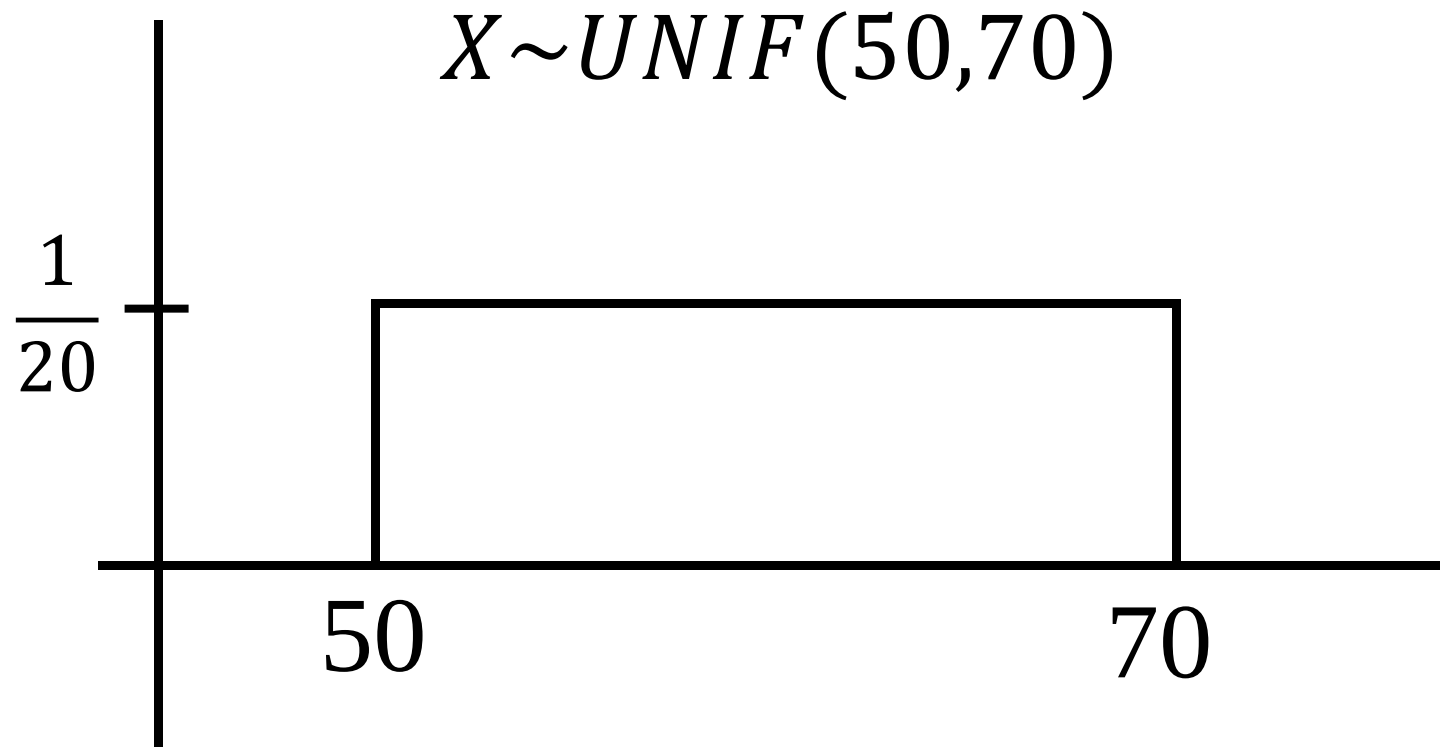
Central Limit Theorem Part 2

Suppose a random sample is taken from a population that isn't normally distributed, with a mean of μ and a standard deviation of σ .

Then the sample mean is approximately normally distributed if the sample size is sufficiently large ($n \geq 30$).

$$\bar{X} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

example: vehicles traveling on a busy freeway have speeds (mph) independently uniformly distributed on the interval (50,70). If a random sample of $n = 50$ vehicles is randomly selected, approximate the probability the mean speed is greater than 59 and less than 61.



Recall the mean, variance, and standard deviation of Uniform.

$$\mu = \frac{\theta_1 + \theta_2}{2} = \frac{50 + 70}{2} = 60$$

$$\sigma^2 = \text{Var}(X) = \frac{(\theta_2 - \theta_1)^2}{12} = \frac{(70 - 50)^2}{12} = \frac{100}{3} \approx 33.3333$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}} \approx 5.7735$$

$$X \sim UNIF(50, 70)$$

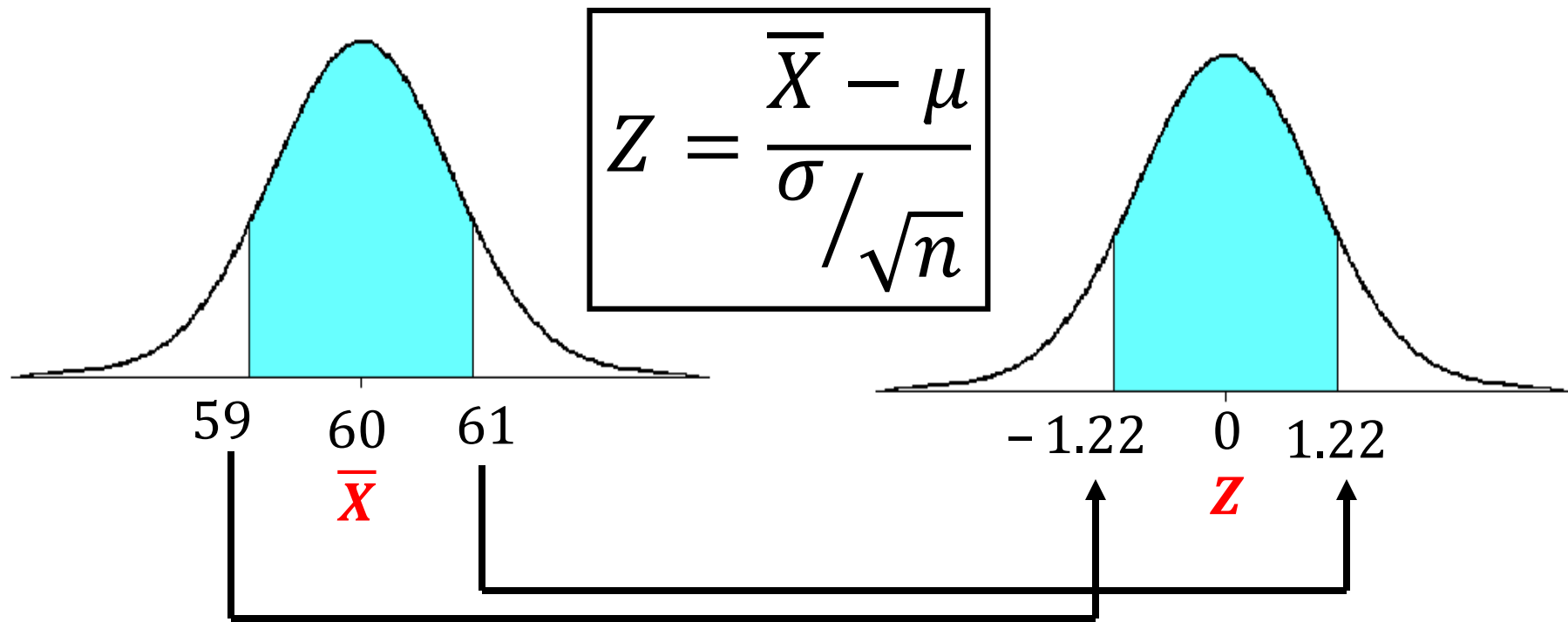
$$\mu = 60$$

$$\sigma^2 \approx 33.3333 \quad \sigma \approx 5.7735$$

If you select $n = 50$ vehicles at random and compute the sample mean, find $P(59 < \bar{X} < 61)$

Since $n \geq 30$ the sample mean is approximately normally distributed.

$$\bar{X} \approx N\left(\mu = 60, \frac{\sigma}{\sqrt{n}} = \frac{5.7735}{\sqrt{50}}\right)$$



$$P(59 < \bar{X} < 61) = P\left(\frac{59 - 60}{5.7735 / \sqrt{50}} < Z < \frac{61 - 60}{5.7735 / \sqrt{50}}\right)$$

$$P(-1.22 < Z < 1.22) = .8888 - .1112 = \boxed{.7776}$$

$$P(59 < \bar{X} < 61) = .7776$$


$$\boxed{n = 50}$$

We just determined the probability of the sample mean being within 1 mph of the mean to equal .7776, if the sample size is 50.

What if the sample size had been 80? Do you think the answer would be larger, smaller, or the same?

$$P(59 < \bar{X} < 61) = \text{????}$$


$$\boxed{n = 80}$$

$$P(59 < \bar{X} < 61) = P(-1.55 < Z < 1.55) = .9394 - .0606 = \boxed{.8788}$$

